

Introduction to Unity’s 2D Environment

The Importance of Concept and Planning Ahead

The most important aspect of a video game is to be *fun* to play. Although graphics and a deep story go a long way, many games that are not particularly strong in those 2 aspects are still successful because they're entertaining.

1) Before you start on your project, sit down and think about what you want your game to be:
   - A 'minutes to kill' or puzzle-solver? (e.g. Angry Birds, Tetris, Sudoku)
   - A classic platformer? (e.g. Super Mario Bros, DuckTales Remastered, Rayman)
   - An adventure/visual novel? (e.g. Detroit: BH)
   - A combination of 2 or more of the above? (e.g. Legend of Zelda, Professor Layton)

The Importance of Story/Premise and Planning (Cont.)

2)After you've decided on a genre, use your favorite games as inspiration

- What are the things your favorite games did GREAT? Re-use the successful concepts and add your own innovations!
- What are the things you didn't like about those games? Bad controls? Ugly backgrounds? Uninteresting story? Recognize their mistakes
- Every work of art is inspired by a work of art that came before it. Do not be afraid to take ideas and lessons from your favorite games. Recycle ideas, but don't plagiarize.

The Importance of Story/Premise and Planning (Cont.)

3) READ about game design. Learn the rules that make a game worth playing. Only when you know the rules, you can experiment by breaking them.

4) Plan your game before you start implementing it. You can make changes while you work, but make sure you at least have a basic map to follow.

What Programs Will We Be Using?

- Unity (Game Engine)
  - Create account then download from the official website
- Visual Studio Code (Script Editor)
  - Download from the Microsoft Store
- Github (Version Control and Collaborative Work)
  - Create account on Github.com, then download *GitBash* to be able to commit and push/pull your work directly from/to your PC

What is Unity?

Unity is a cross-platform game engine with a built-in IDE developed by Unity Technologies. It is used to develop 2D and 3D videogames for web plugins, desktop platforms, game consoles, and mobile devices.

– *Wikipedia*

What is IDE, and what is a Game Engine?

- **IDE** stands for “integrated development environment”.
  - An example of an IDE you already know is Netbeans
  - IDE is a software application that provide comprehensive features to computer programs for software development. An IDE brings many of those development-related tools together as a single framework, rather than have the developer select and manage all of these tools separately.

- A **game engine** allows a game created to run in different environments
  - A game engine is framework for game development. This framework helps with several core areas that all games have:
  - Graphics
  - Audio
  - Logic

What is framework?

- It is an abstraction in which common code providing generic functionality can be selectively overridden or specialized by user code providing special functionality.

- That’s why the game engine is a framework – it provides tools and structures that every game requires to function.

- Using an engine means you don’t have to reinvent the wheel every time you start a new project.

First Look at Unity

- Download **Unity Hub** from their official website. Run it and sign up with a Unity account. Make sure to remember your username and password for later.

- After that, go to https://unity.com/releases/editor/archive and download Unity version 2021.3.10f1. When you try to install it, it will automatically open Unity Hub to begin the process.

First Look at Unity (Cont.)

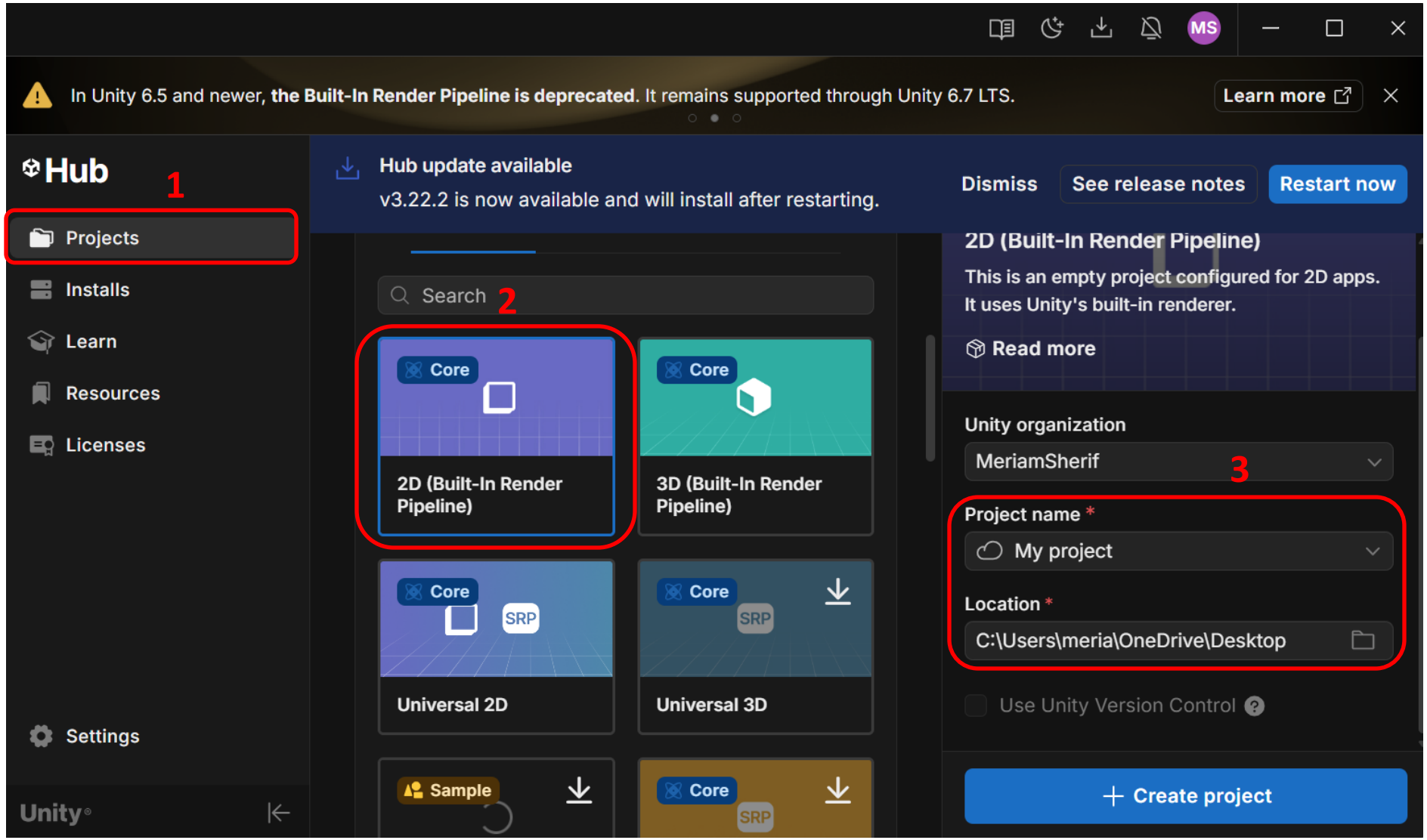
- Now to create a new Project in Unity Hub: go to the Projects tab on the left, and choose **Universal 2D** from the list.

- On the right, give the project a name (e.g. MyFirst2DGame) and choose the path/location where the project will be saved locally on your PC.

- Click Create Project.

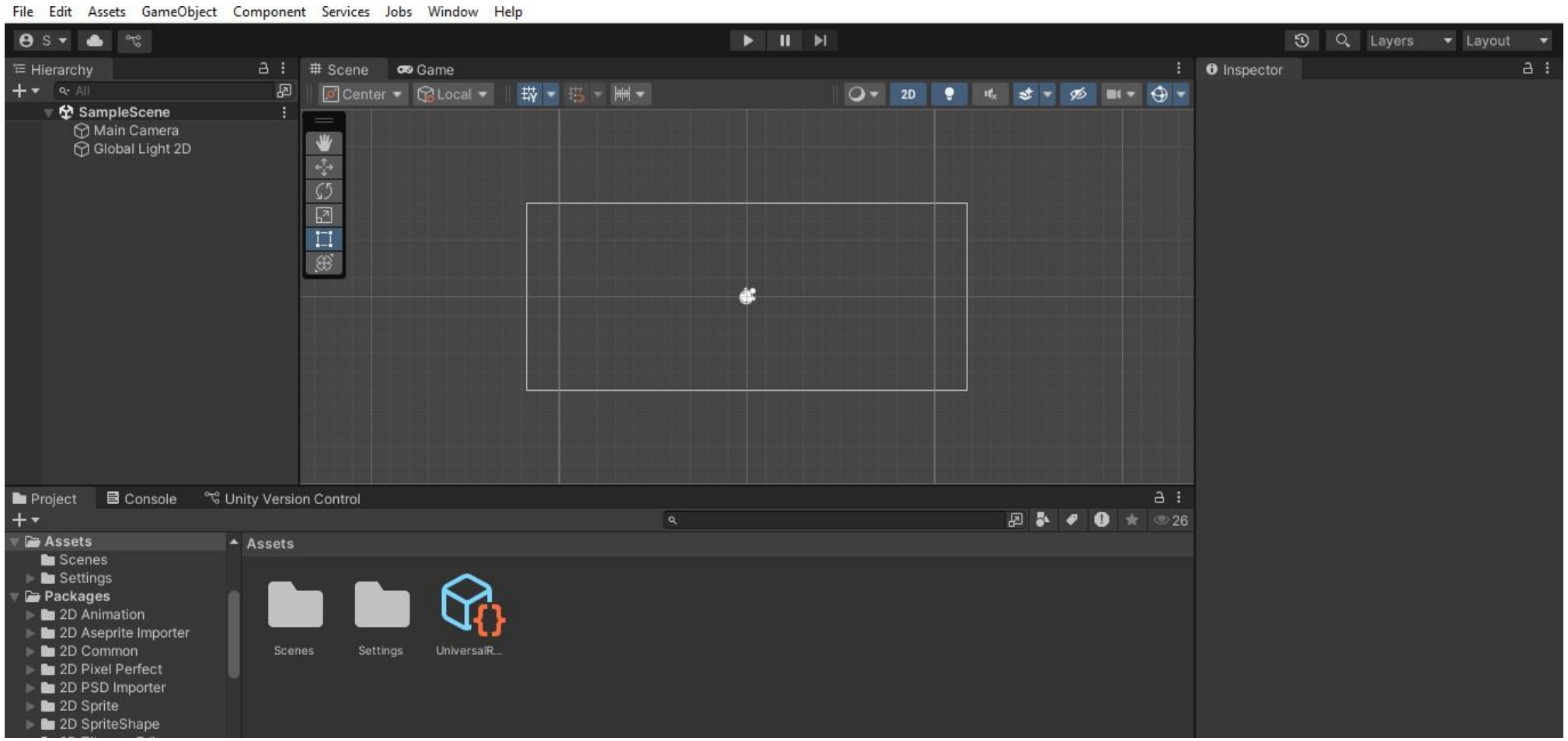
(See next slide for the screenshot example.)

First Look at Unity (Cont.)



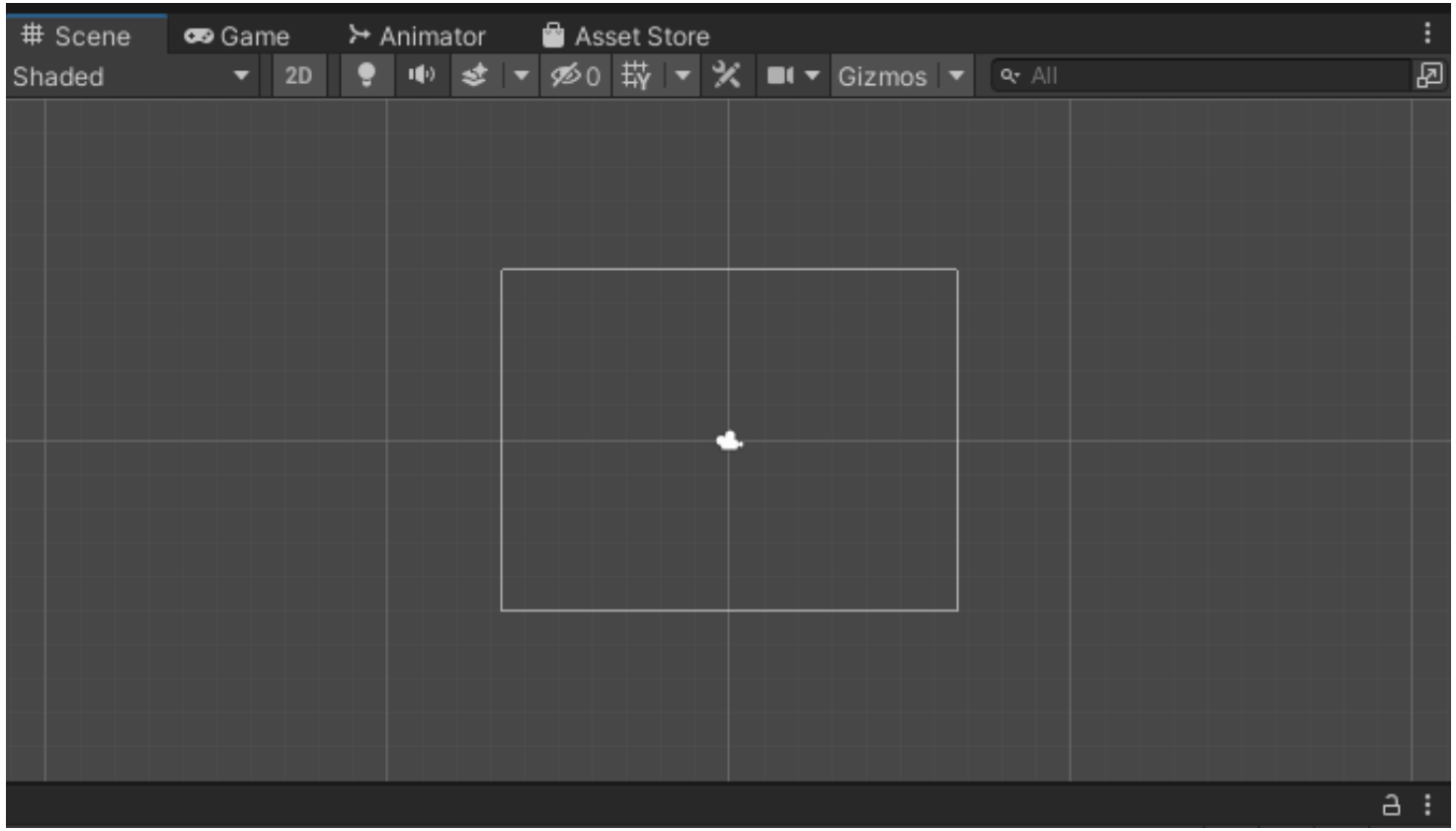
First Look (Cont.)

- This is your default workspace upon opening the project.



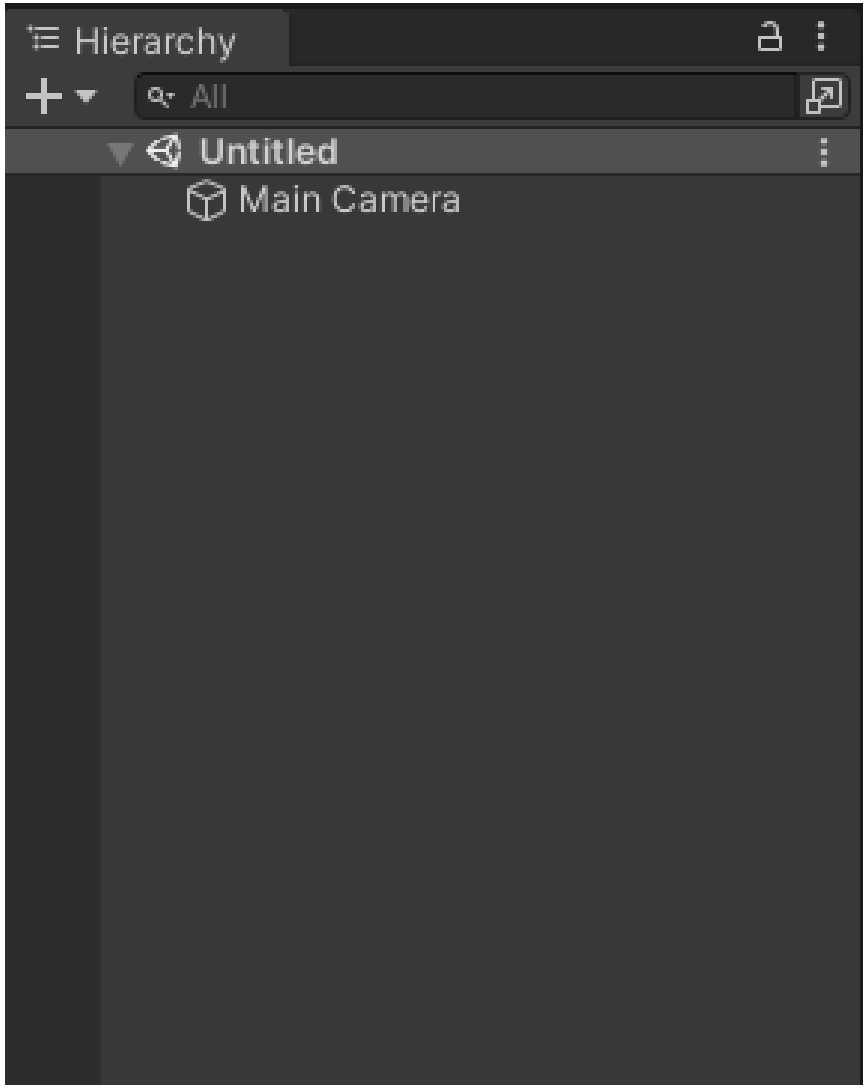
First Look (Cont.)

- Scene window
  - This is where you assemble all the elements of your scenes in the game



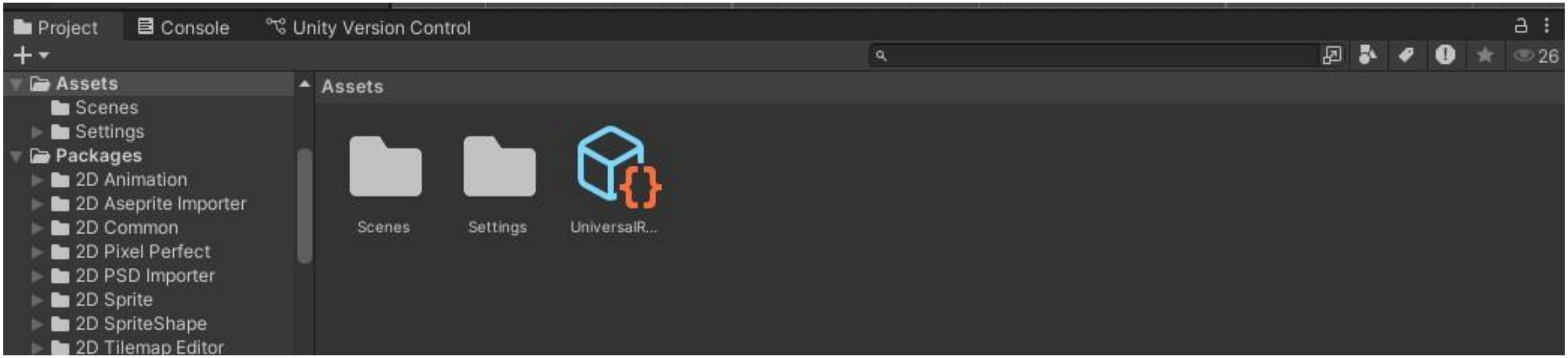
First Look (Cont.)

- Hierarchy window
  - Lists all the elements that exist in your scene, and establishes parent-child relationships between them.



First Look (Cont.)

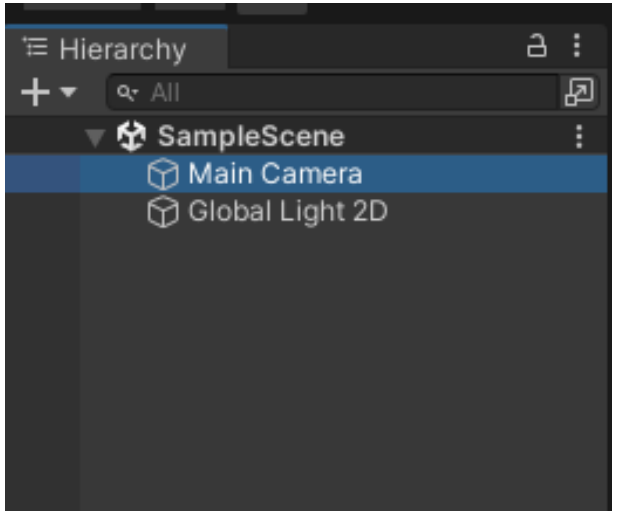
- Project window
  - This is where you organize all the GameObjects that you'll be using in your scene, from character sprite sheets to animation to background imagery to code. Note: keep your assets organized by putting them in folders! (e.g. Scripts, CharacterSprites, etc.)

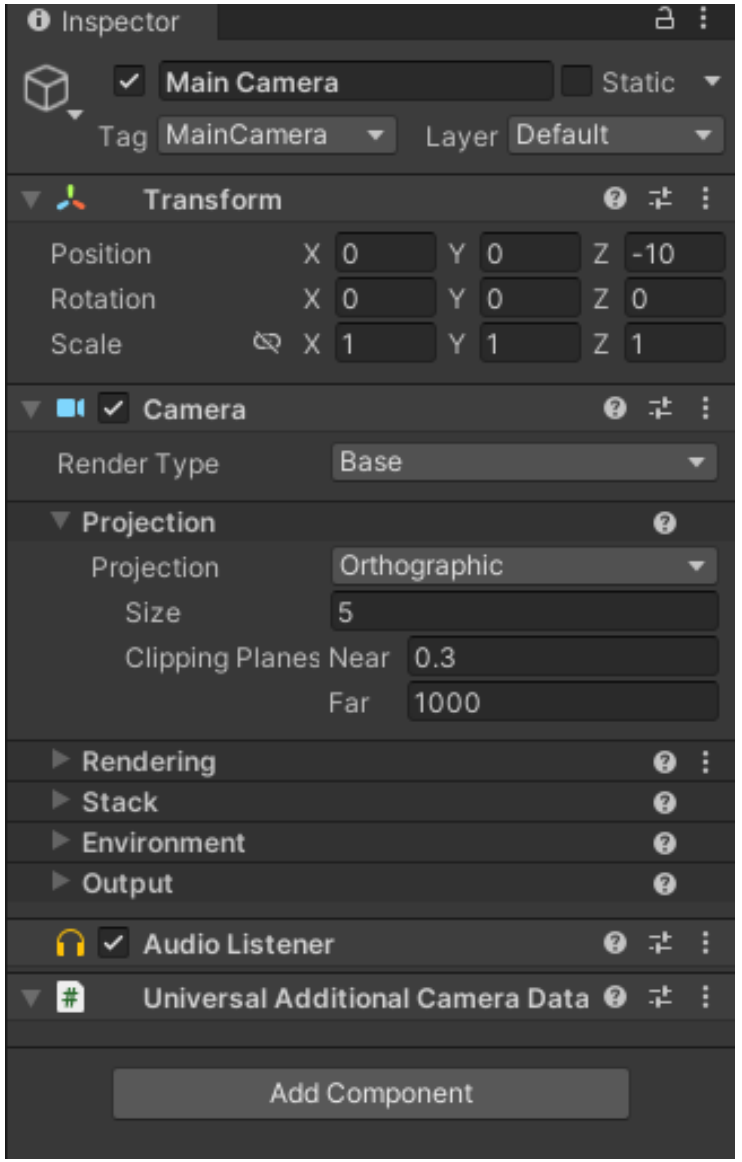


First Look (Cont.)

- Inspector window
  - This is where you can see and edit the properties of any element you've selected to work on.

    For example, if you select the Camera in your scene, you'll see its properties populate the Inspector.





First Look (Cont.)

- Transform tools
  - Easy-access tools to change the position, size and orientation of any element in your scene
  - The hand tool moves the entire scene window canvas
  - The move tool moves a selected element around the scene window
  - The rotate tool rotates a selected element around the x or y axis
  - The transform/scale tool allows for increasing the size of an element along the x axis, y axis or both

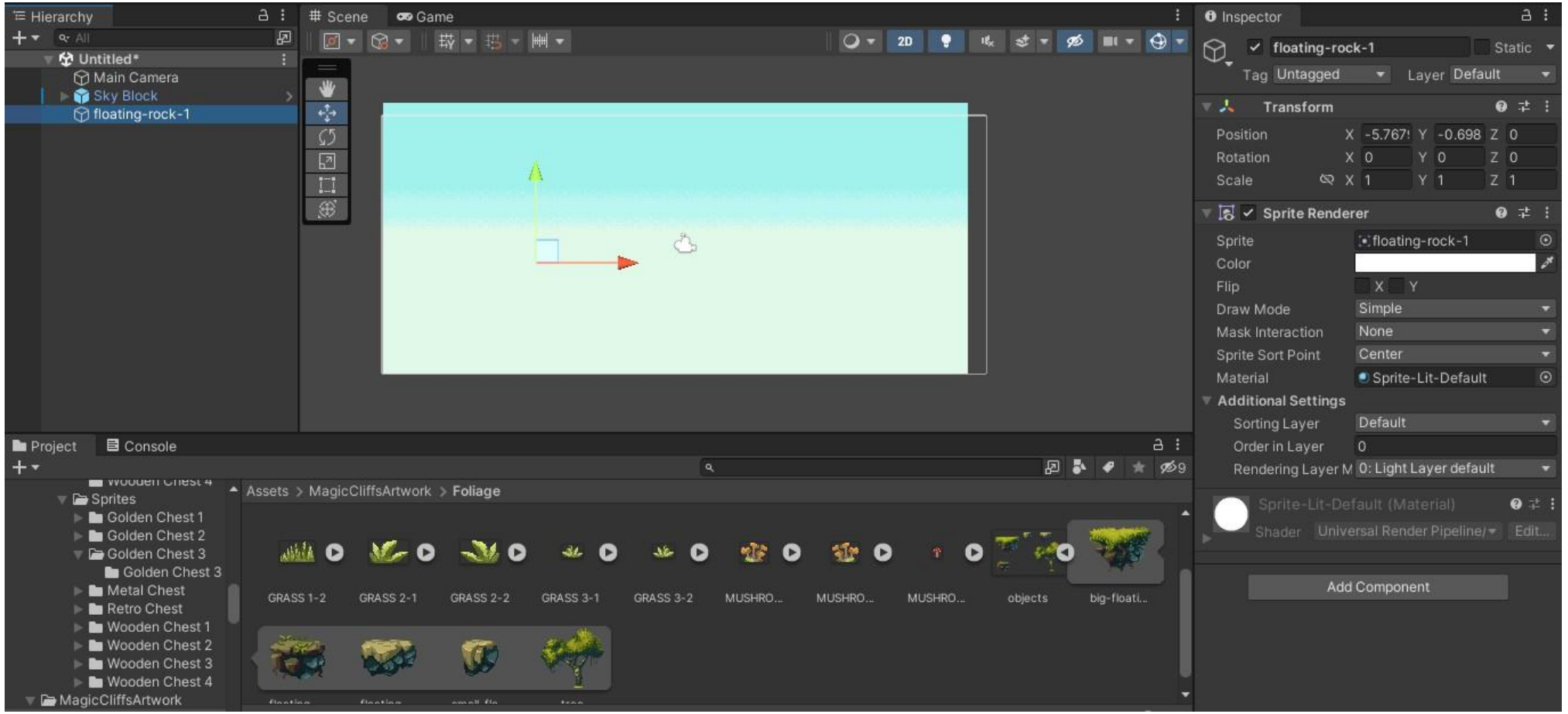


Exercise #01 – Compose a Simple BG

- Go online and choose several elements to compose a pleasant foreground and background. E.g. a character, some trees, grass, mountains, birds, blue sky, etc. Be creative!
- Create a folder in your project window. Name it Background_Sprites.
- Drag and drop sprites into the folder
- Drag and drop the sprites from the folder to the Hierarchy window. They'll automatically also appear in the Scene window. Assemble everything to create an interesting background.
- Use the transform tools in the toolbar to help you rotate, resize, etc.

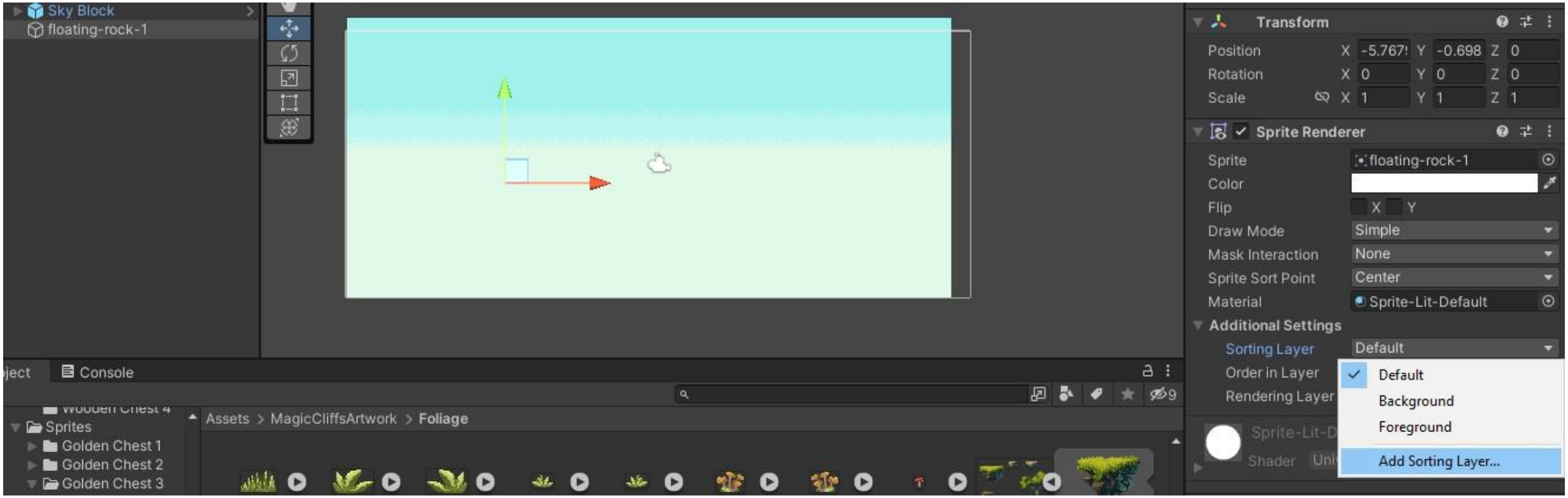
Exercise #01 – Create a Simple BG (Cont.)

- You may find that the images are not being assembled the way you want (e.g. the suspended platform is hiding behind the sky)



Exercise #01 – Create a Simple BG (Cont.)

- To re-assemble them correctly, select the platform and go to the inspector window
- Choose Sorting Layer > Add Sorting Layer



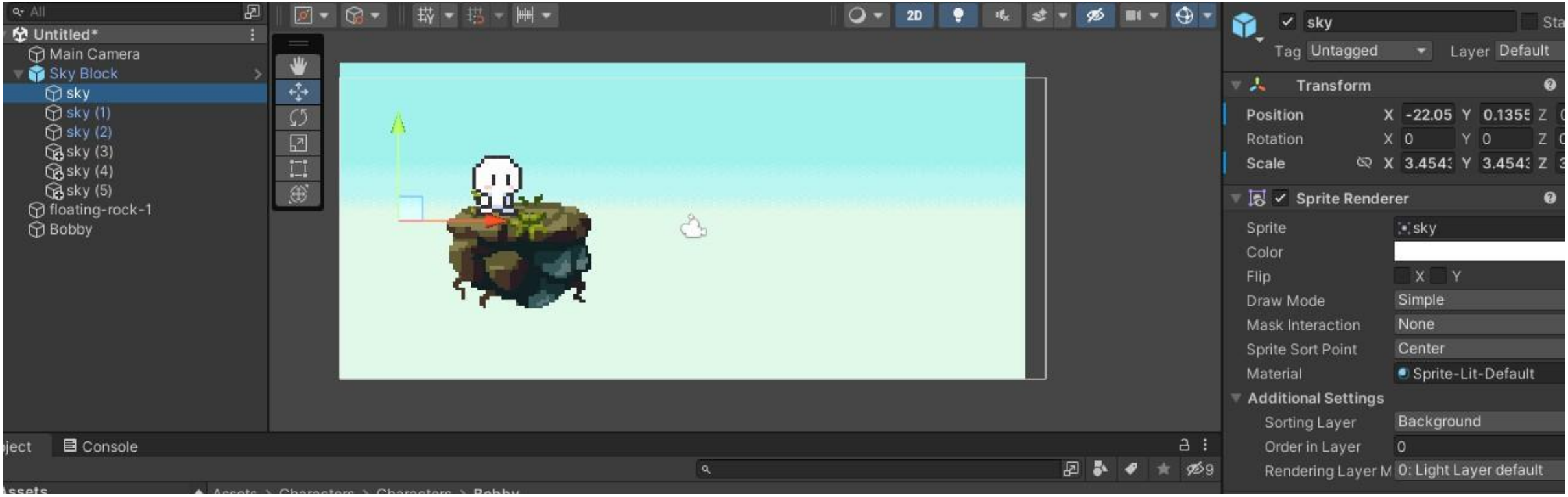
Exercise #01 – Create a Simple BG (Cont.)

- Click the + sign, and add a new layer named **Foreground**. The horizontal lines on the left of each layer allow you to re-assemble if needed



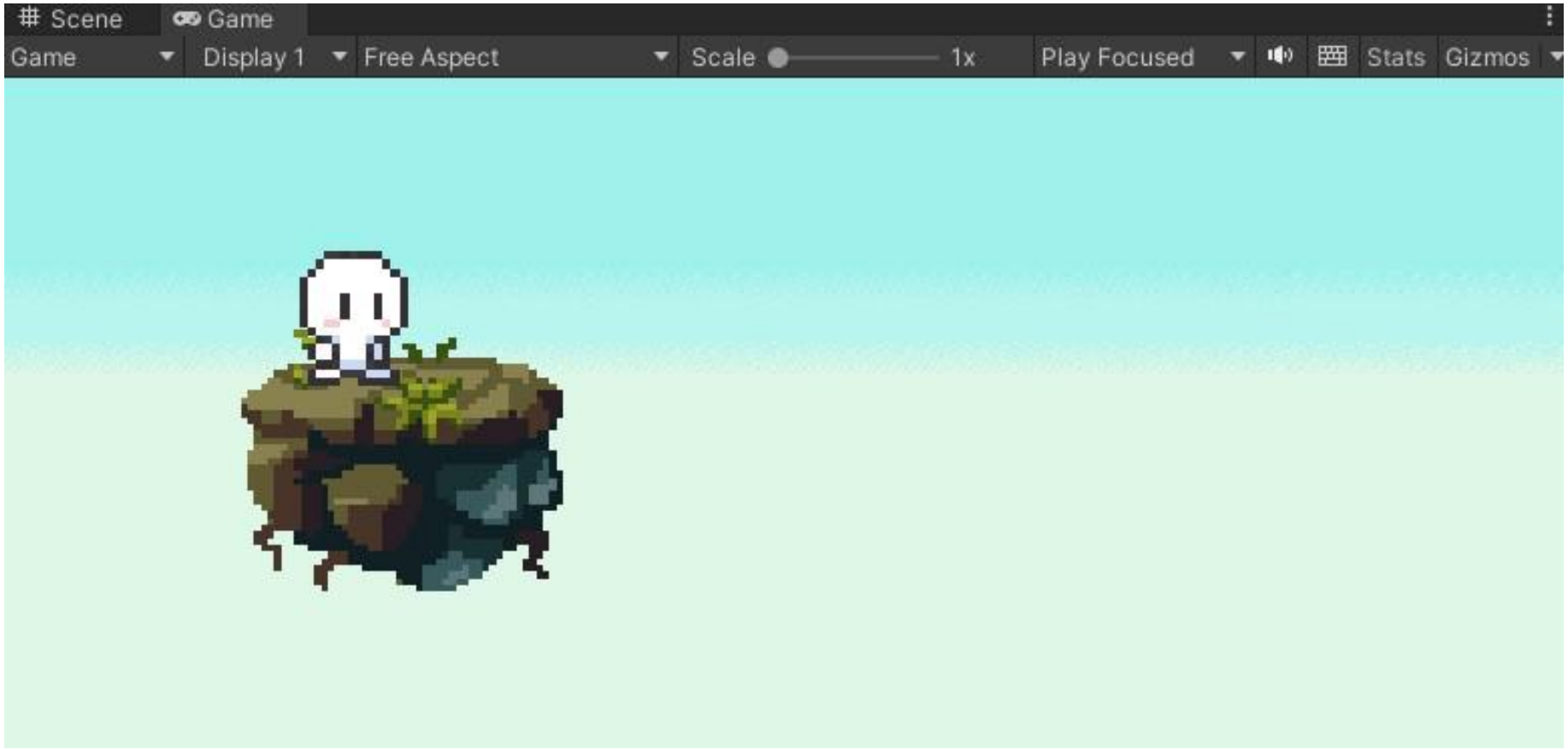
Exercise #01 – Create a Simple BG (Cont.)

- Re-click on the platform's sprite name in the hierarchy window, and change its sorting layer from Default to Foreground
- Now create another sorting layer and name it **Background**. Place all your background elements like the sky in it



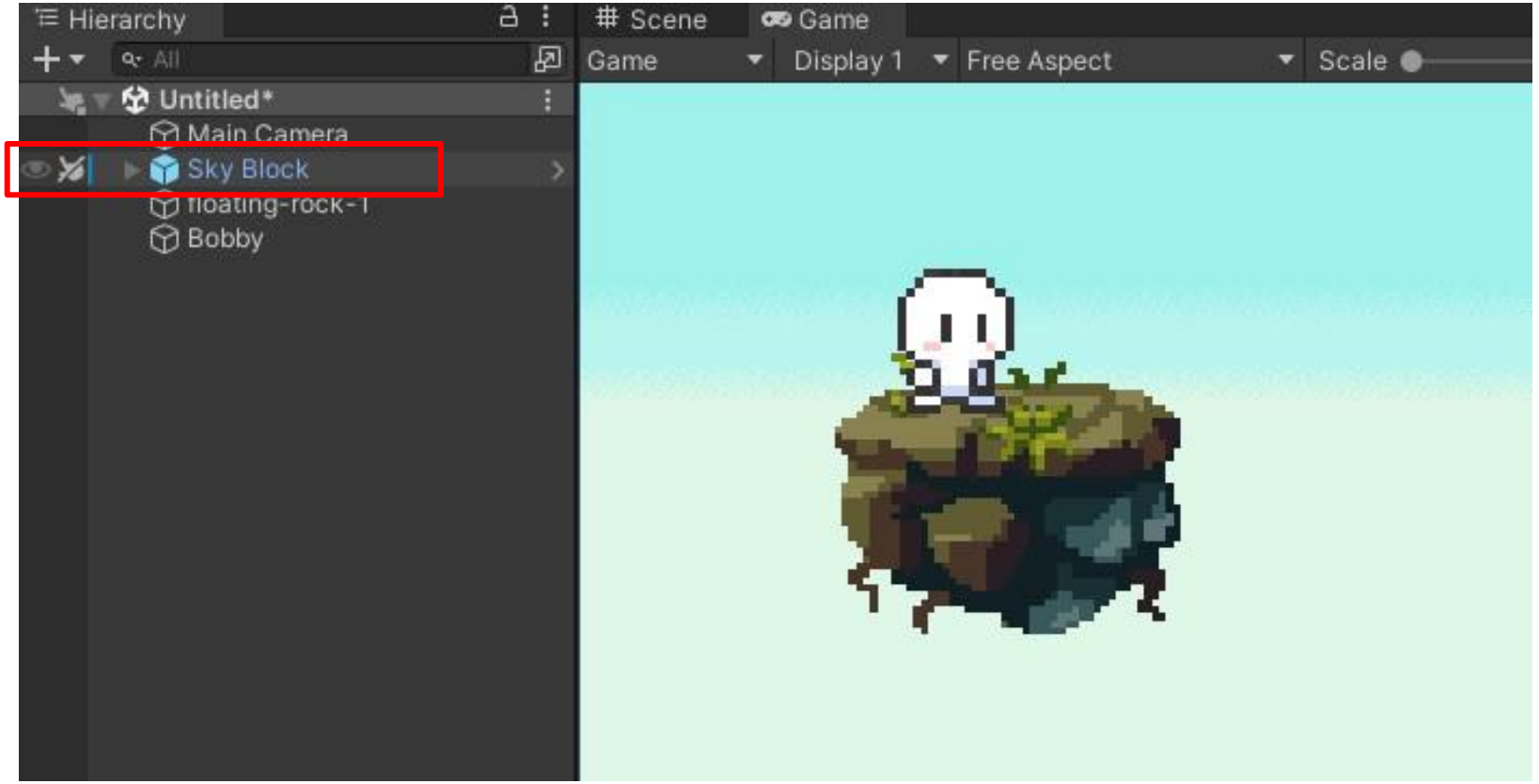
Exercise #01 – Create a Simple BG (Cont.)

- To see what your scene looks like from a player's perspective, click the Game window.



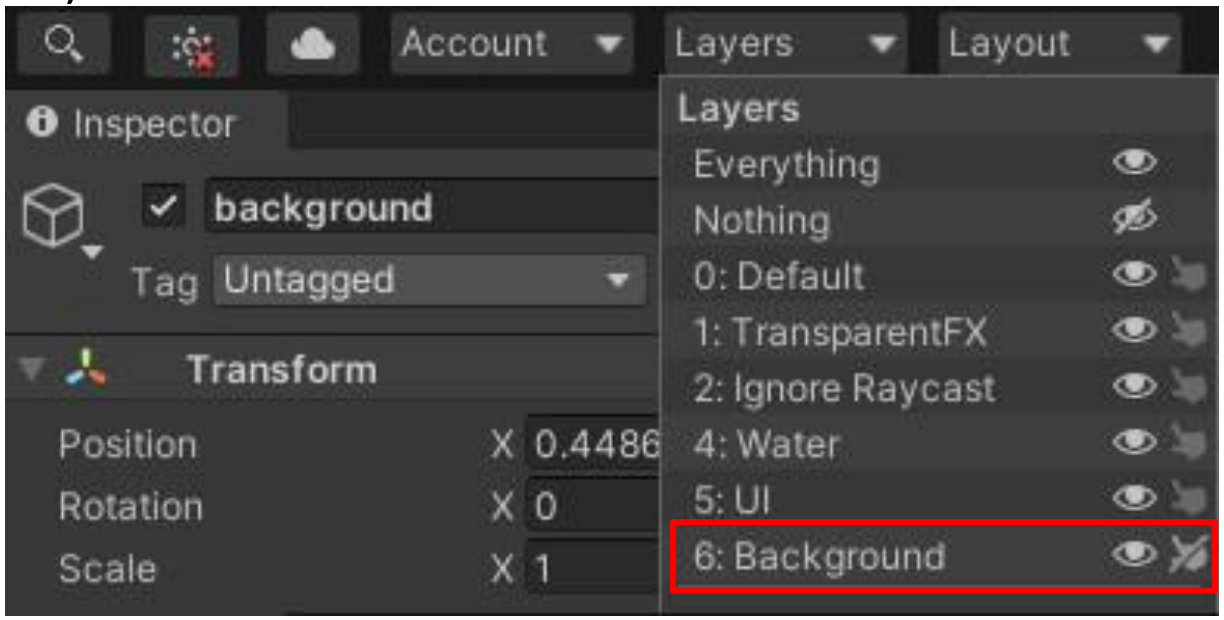
Exercise #01 – Create a Simple BG (Cont.)

- To lock a gameObject and avoid accidentally moving it around, click the pointing hand icon next to it in the Hierarchy (e.g: lock the sky).



Exercise #01 – Create a Simple BG (Cont.)

- And to lock *an entire layer* with everything on it, not just individual gameObjects, go to the Layers drop-down list above the Inspector window and choose Background (or whatever layer you had created).



- To lock the layer, click the pointing hand icon.

- **<u>Quick reminders:</u>**
- *Sorting Layer* helps organize how sprites are seen by the camera. Since it is a property of the Sprite Renderer component, it determines how sprites are rendered in the scene.

- As for *Layer*, which is above the Inspector window next to Tags, is for grouping objects and can be later manipulated in code. For example, you may want some layers to allow collision (e.g. Ground), but not others.

- So Sorting layers serve the *visual* side, whilst Layers more serve the *functional* side.

Your Project

- Your project in this module is to create an intuitive 2D platformer game. It does not have to be complex or long. Creating an entertaining retro game with **smooth gameplay** and a **clear story** is completely acceptable.



Reminder

- Before creating your level, determine the type/genre of game you want to make.
- Plan all the important stuff on paper. Draw sketches.
- Determine the aesthetic and atmosphere, because they will dictate your art style and music.

Useful References

- **Game Conceptualization and Design:**
  - Johnson, M., Hasankolli, R., & Henley, J. A. (2014). *Learning 2D Game Development with Unity: A Hands-on Guide to Game Creation*. Pearson Education
  - **The Big List of Game Design. URL retrieved from:** http://www.pixelprospector.com/the-big-list-of-game-design/
  - **Jonkers, D. (2011). How to design Levels for a Platformer. URL retrieved from:** http://devmag.org.za/2011/07/04/how-to-design-levels-for-a-platformer/
  - **Game Background Art on Pinterest. URL retrieved from:** https://www.pinterest.com/explore/game-background-art/
  - **How to design levels for a platformer. URL retrieved from:** http://devmag.org.za/2011/07/04/how-to-design-levels-for-a-platformer/
  - **11 Tips for making a fun platformer. URL retrieved from:** http://devmag.org.za/2011/01/18/11-tips-for-making-a-fun-platformer/

Useful References

- **Level Analysis and Design:**
  - Johnson, M., Hasankolli, R., & Henley, J. A. (2014). *Learning 2D Game Development with Unity: A Hands-on Guide to Game Creation*. Pearson Education.
  - **How to make a 2D Platformer – Unity Tutorial by Brackeys. URL retrieved from:** https://www.youtube.com/playlist?list=PLPV2KyIb3jR42oVBU6K2DIL6Y22Ry9J1c
  - **Beginning Level Design by Tim Ryan. URL retrieved from:** http://www.gamasutra.com/view/feature/131736/beginning_level_design_part_1.php?page=1
  - **Becoming a Level Designer and Env. Artist Part 1.URL retrieved from:** http://www.worldofleveldesign.com/categories/level_design_tutorials/becoming-level-designer-environment-artist-part1.php
  - **The Level Design Analysis Playlist by Sunder. URL retrieved from :** https://www.youtube.com/playlist?list=PLHKJTKyEfjmGegraWzCJYjbJYnIbOHi-y
  - **100 Level design Ideas and Locations. URL retrieved from :** http://www.worldofleveldesign.com/categories/level_design_tutorials/100_level_design_ideas/100_level_design_ideas_and_locations.php